

Exposome and covariate impact on child neurobehavioral functioning

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1 Introduction

The HELIX study represents a collaborative project across six European countries (France, Greece, Lithuania, Norway, Spain, and the United Kingdom). The aim of this study is to analyze the effects of the exposome on the phenotype of children from 6 to 11 years old.

In this report, we focus on the effects of the exposome on their **neural behavior** (**'hs_Gen_Tot'**), measured with the CBCL scale, by carrying out a statistical study on the data provided by the ATHLETE project (Simulated data, based on the HELIX project, publicly available to challenge researchers on statistical tools to study exposome-health associations).

Firstly, we outline the methods employed, detailing the procedures, tools, and techniques utilized. Then, we present the results obtained. And finally, we discuss these results.

2 Methods

2.1 Study participants

We carry out our study on all six cohorts of the data provided : France, Greece, Lithuania, Norway, Spain, and the United Kingdom. The study population consists of 1301 mother-child pairs where biomarkers, omics signatures and child health outcomes were measured at age 6-11 years.

2.2 Statistical analysis

We used several methods [3] to simplify our model and to keep only the most significant variables.

2.2.1 Pre-processing

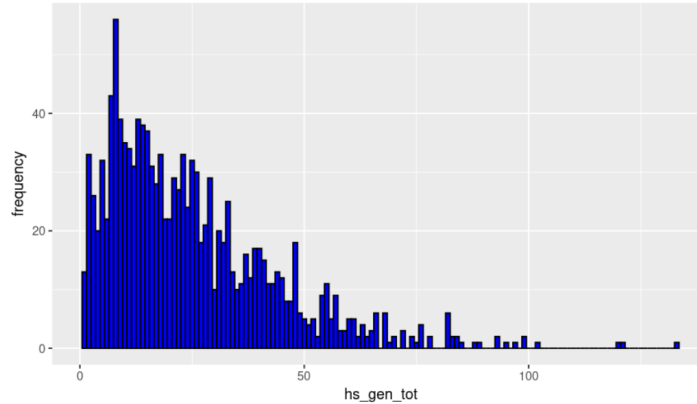


Figure 1: hs_Gen_Tot distribution

First, by taking the logarithm of our variable of interest (**hs_Gen_Tot**), we changed it from its initial distribution to a **gaussian distribution**. This step ensures that we can then apply regression methods that require gaussian-distributed values.

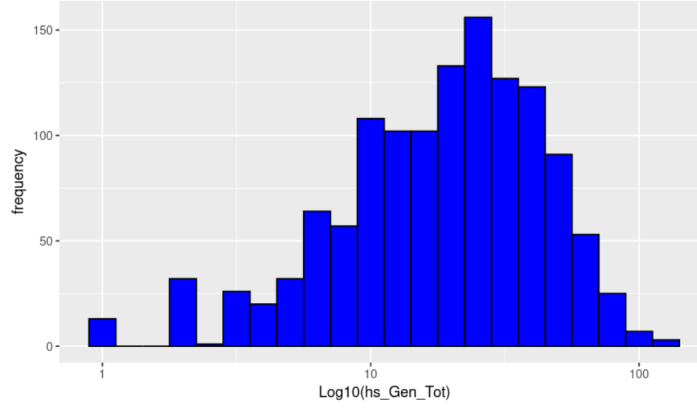


Figure 2: Log₁₀ hs_Gen_Tot distribution

2.2.2 Principal component analysis

Second, we coarsely explore the data provided with a **PCA** on the **exposome**, in order to better understand and represent the influence they have on exposomes and phenotypes, and thus the bias they may produce.

2.2.3 Selection of covariates

Covariates are variables that affect a response variable, but are not of interest in a study. We start by evaluating the p-values of each model taking into account a **single covariate** to thus determine the most relevant ones. As is customary, a threshold of **0.05** was chosen for the selection of relevant covariates

We also perform this analysis by examining **all covariates** at once with the stepwise method, to ensure that any potential combinations of covariates which may not be significant individually, are considered for their potential **combined relevance**.

Finally we analyzed the correlation between the different covariates and only kept the **independent** ones to simplify our model and ensure that it is not biased. To select them, we used a heat map.

2.2.4 Selection of relevant variables of exposome

After selecting the most significant covariates we look at the influence of the **individual variables** of the exposome on **hs_Gen_Tot**, with a monivariate and multivariate regression (Lasso). The Group Lasso method was then used to take into account intrinsic classification of variables, including **families, subfamilies** and **period (prenatal or postnatal)**.

3 Results

3.1 Principal component analysis

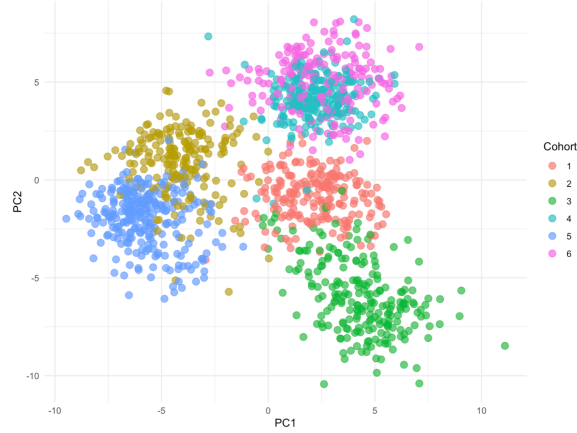


Figure 3: PCA

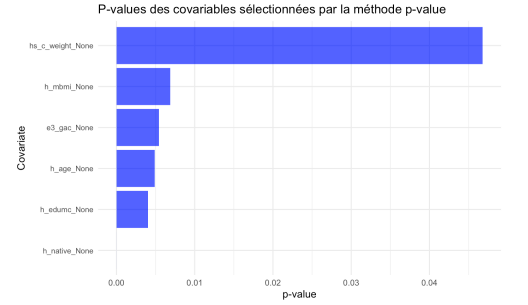
The PCA highlights the effect of cohorts on phenotypes. Though the principal components are not explicitly stated here, this visualization proves the importance of taking into account the effect of cohorts on our variables, in order to carry out an unbiased study.

3.2 Selection of covariates

Covariates were firstly selected through p-value selection, of which the results can be found hereafter:

Covariate <chr>	p_value <dbl>	Type <chr>
h_mbmi_None	6.872596e-03	P-value
e3_gac_None	5.428150e-03	P-value
h_age_None	4.906808e-03	P-value
h_edumc_None	4.054112e-03	P-value
h_native_None	3.487736e-06	P-value
hs_c_weight_None	4.680843e-02	P-value

(a) Covariates with p-values under 0.05



(b) Histogram of covariate p-value distribution

Figure 4: Comparison of covariate p-values and their distribution

The stepwise method was also used in order to mitigate cocktail effects which may be present on covariates :

Covariate <chr>	Type <chr>
h_cohort	Stepwise
e3_gac_None	Stepwise
h_age_None	Stepwise
h_edumc_None	Stepwise
h_native_None	Stepwise
hs_c_height_None	Stepwise
hs_c_weight_None	Stepwise

Figure 5: Covariates selected through stepwise method

Finally, the covariable heatmap allowed for a more global understanding of how covariates interact with each other, and take into account epigenetic considerations.

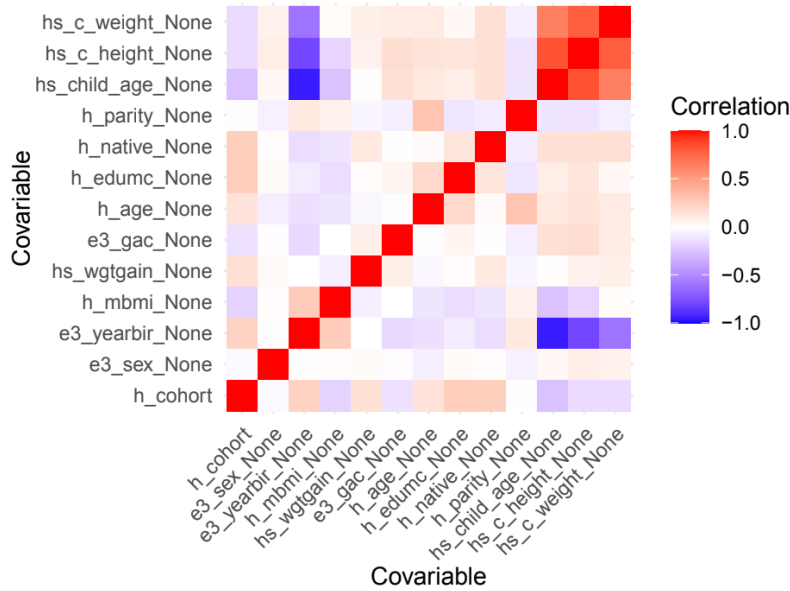


Figure 6: Heatmap of the correlation of the covariates

The heatmap, coupled with the stepwise and p-value studies, as well as epigenetic considerations, lead us to only take into account for the rest of the study : **h_cohort** (Cohort of inclusion), **h_native_None** (If the parents are native from the country of the cohort), **e3_sex_None** (Child sex, female or male), **h_mbmi_None** (Maternal pre-pregnancy body mass index), **e3_gac_None** (Gestational age at birth in weeks) and **hs_c_weight_None** (Weight of the child at 6-11 years old).

3.3 Selection of relevant variables of exposome

3.3.1 Monivariate method

In accordance with announced

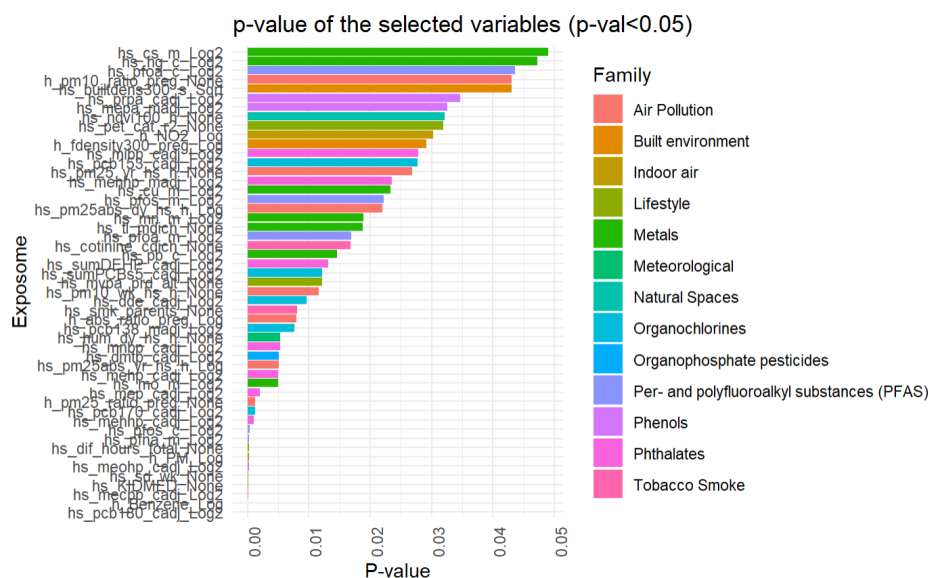


Figure 7: Exposome's most significant families - monovariate regression

Out of 222 variables of exposome, we kept these 51 most significant ones based on the lowest p-values (<0.05). Four out of thirteen families make up for around 50% of the impacting exposome : **Phthalates**, **Air Pollution**, **Metals** and **Organochlorines**. The families having the most representative exposomes were then put forth through the following figure :

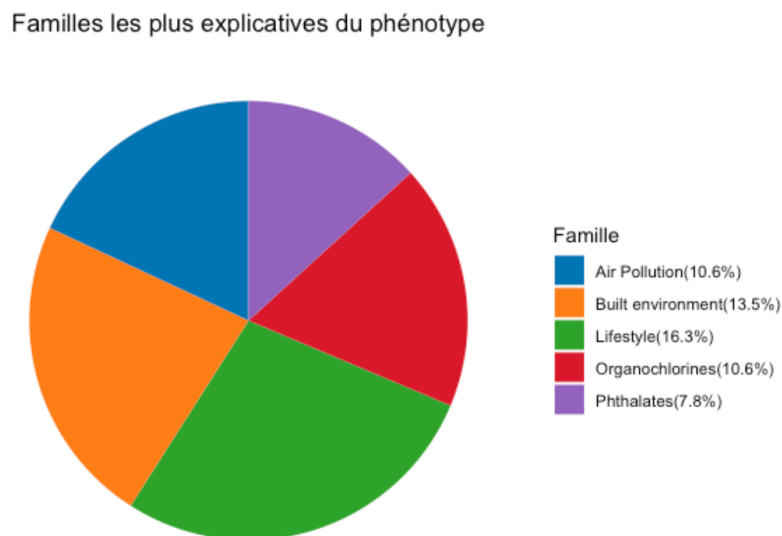
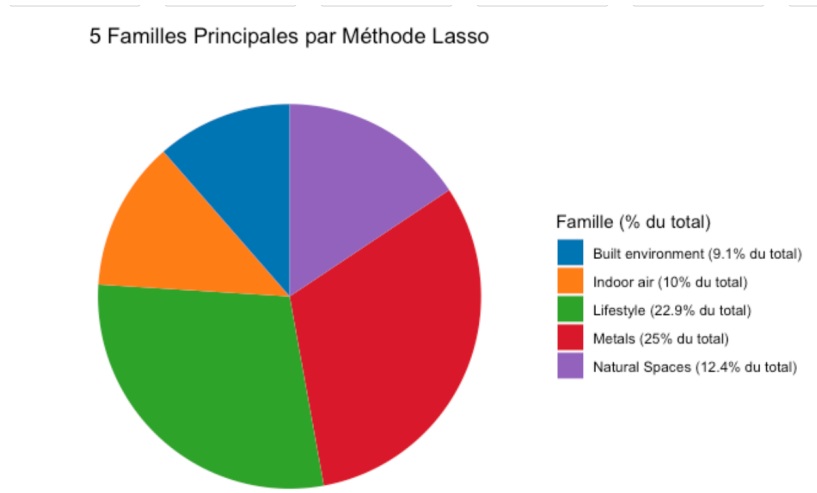


Figure 8: Exposome's most significant families - monovariate regression

Since this method relied solely on p-values to indicate correlation between exposomes and neural behaviour, indicators such as RMSE or R^2 are not relevant to the study, but will be taken into account in further methods. The preliminary conclusions here are thus that **Lifestyle**, followed by **Built Environment** and **Air Pollution** have the most significant impact of child neural behaviour.

3.3.2 Lasso Method

The next method taken into consideration in order to control the contribution of all exposomes at once, rather than taking exposomes one by one, is the Lasso method. The method allows us to better analyze potential cocktail effects which would have not appeared through the monovariate method. The results of the method, presented in the same way as the monovariate method, with the **five most representative families**, are as follows :



(a) Most representative families according to LASSO method

R_squared<dbl>	RMSE<dbl>	MAE<dbl>
0.2675615	0.7674873	0.5903055

(b) LASSO method quantitative metrics

Figure 9: LASSO method results

The accuracy of the LASSO method, with an **R squared** of around **0.27** is rather satisfying as the study is necessarily capped by the fact that all factors relative to neural development are not taken into account in the study, such as IQ, or whether the child goes to school, and so on... It can be interpreted that the LASSO model makes up **26 percent** of the neural behaviour.

3.3.3 Group Lasso Method

The last method taken into account was the Group LASSO method, in order to take into account the intrinsic classification of the different exposomes due to their nature. The algorithm for such a method however, is very time consuming, the sample had to be **reduced from 1301 samples to 225 samples**, above 225 samples, the algorithm would not give any results. The results are as follows :

Familles principales (Group Lasso) pour un échantillon de 225

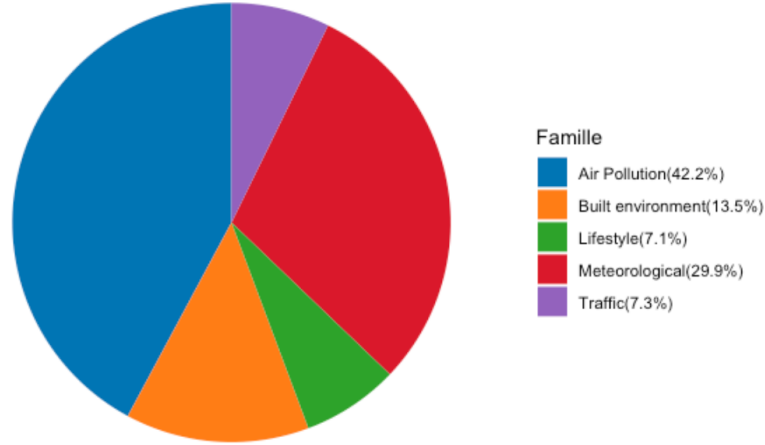


Figure 10: Most Significant Families - Group LASSO method

The precision metrics are irrelevant in this case due to the size of the sample. With enough computational power, it would have been interesting to pursue this method and implement in on the entirety of the available data for more relevant results.

3.4 Results comparison

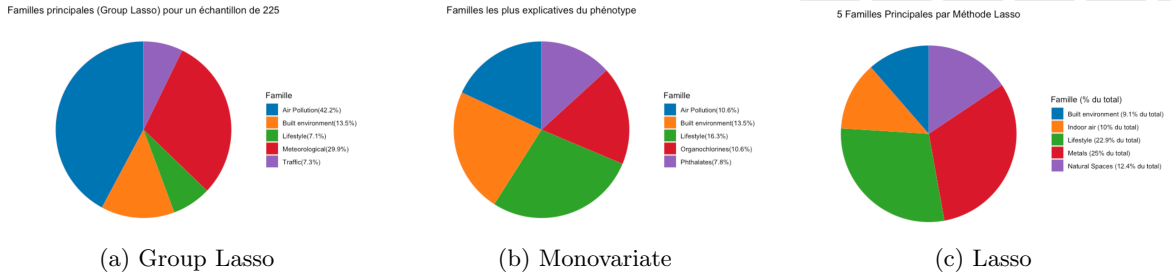


Figure 11: Comparison of Main Families Identified by Different Methods

Commentary on Differences

The three figures above illustrate the main families identified by the Group Lasso, Monovariate, and Lasso methods, respectively.

- **Group Lasso:** This method identified **Air Pollution** as the most significant family, accounting for 42.2% of the total exposomes. The **Built Environment** and **Meteorological** families also showed considerable importance.

- **Monovariate:** In contrast, the Monovariate method highlighted **Lifestyle** as the most explanatory family for the phenotype, with a share of 16.3%. This method provides a different perspective, showing a **more even distribution** across various families.

- **Lasso:** The Lasso method indicated **Metals** as the most significant family, contributing 25% to the total. This method's results differ markedly from the Group Lasso and Monovariate methods, emphasizing different aspects of the data.

3.5 Lasso Method Deep-dive

Out of the **three** methods put forth, the LASSO method seemed most promising due to its **ease of use, low time complexity and precision**. Further analysis were made using this same method in order to better understand how the main families affect neural behaviour.

To this extent, **subfamilies** were taken into account. For the three main families, the LASSO method was used to determine the main subfamilies that explain the phenotype within the family. Thus, according to previous results, the families taken into account are : **Metals** (25% of total exposomes), **Lifestyle** (22.9% of total exposomes) and **Natural Spaces** (12.4% of total exposomes). The results of the study are as follow :

Subfamily Analysis

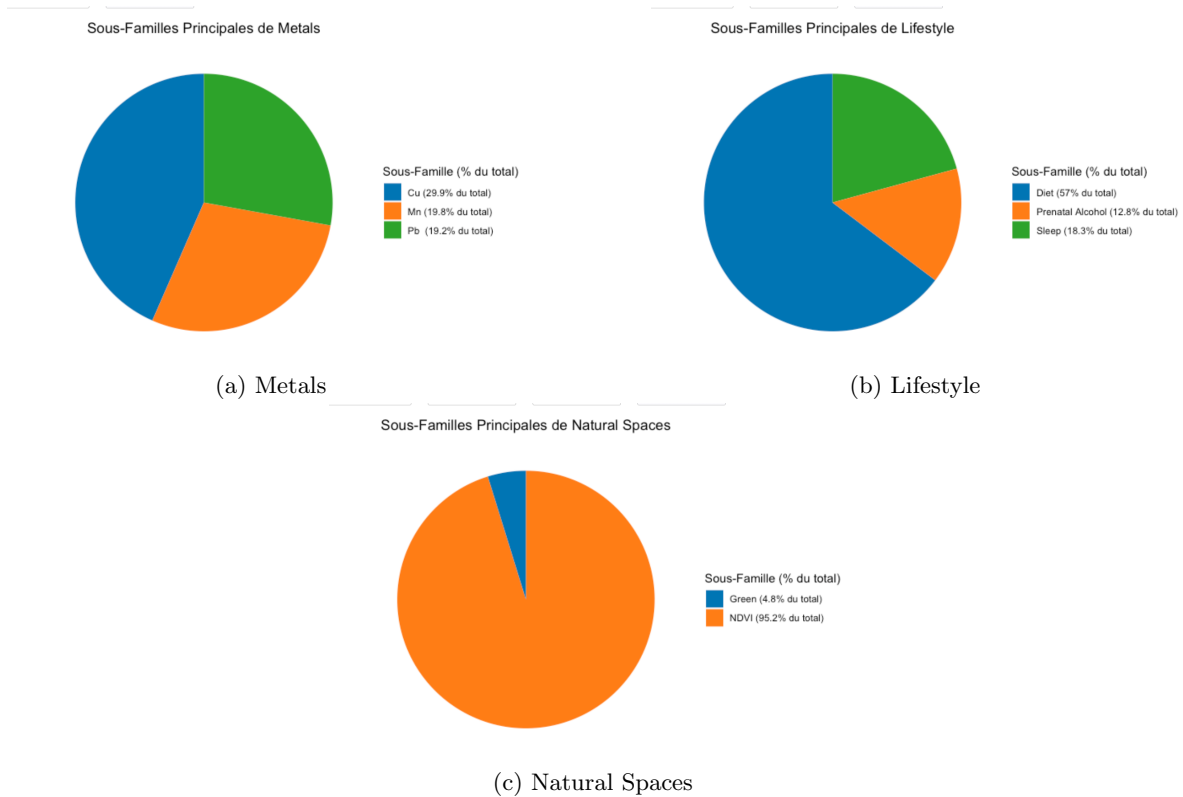


Figure 12: Sub-Families Comparison

Commentary on Contributions

The three pie charts above illustrate the contribution of main families to the phenotype according to the Lasso method.

- **Metals:** The Metals family shows a more balanced contribution among its sub-categories. **Copper (Cu)** contributes the most with 29.9%, followed by **Manganese (Mn)** with 19.8%, and **Lead (Pb)** with 19.2%.

- **Lifestyle:** The Lifestyle family is predominantly represented by **Diet**, contributing 57% to the total. **Prenatal Alcohol** and **Sleep** are also notable, accounting for 12.8% and 18.3%, respectively.

- **Natural Spaces:** In the Natural Spaces family, **NDVI** (Normalized Difference Vegetation Index) overwhelmingly dominates with 95.2%, while **Green spaces** account for only 4.8%.

Period Analysis

A similar study was carried out on the periods most representative of each families, in order to understand, for later conclusions, if the effects are due to **prenatal** or **postnatal** elements of each given family. Results are as follow :

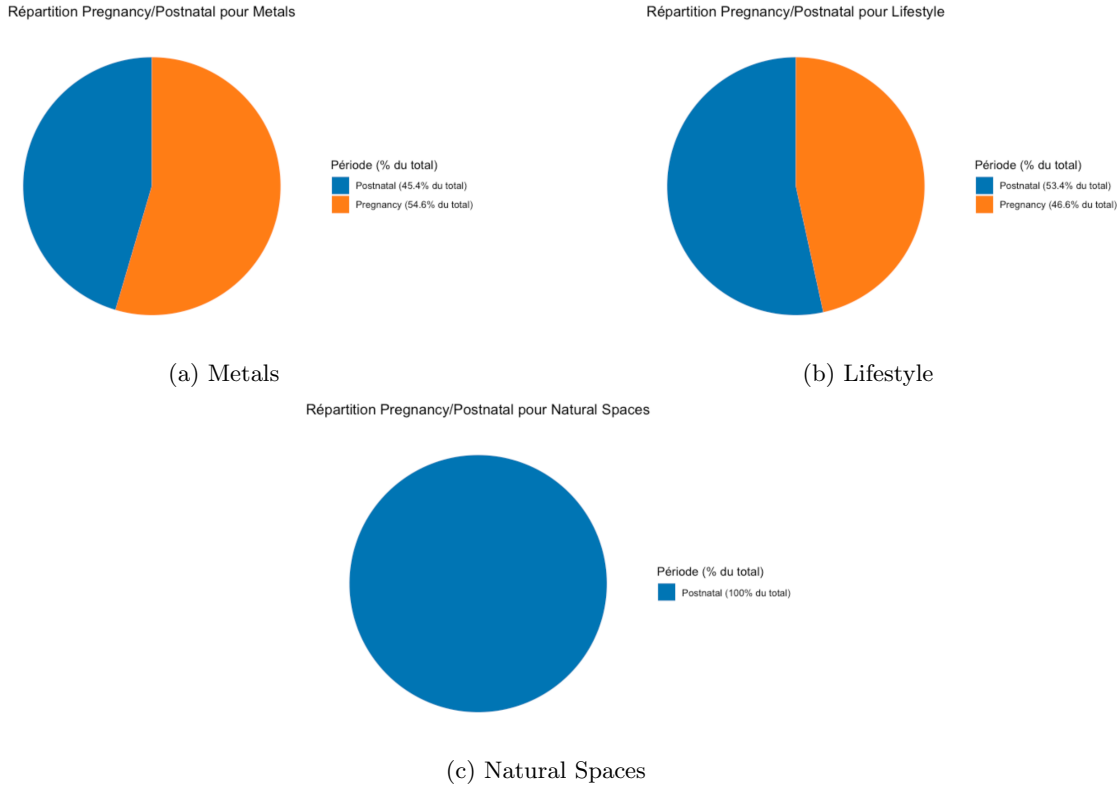


Figure 13: Comparison of Pregnancy/Postnatal Distribution for the three main families

Commentary on Prenatal and Postnatal Distribution

The three pie charts above illustrate the prenatal and postnatal distribution for the main families.

- **Metals:** The Metals family presents a slightly predominant influence during the prenatal period, with 54.6% compared to 45.4% postnatally.

- **Lifestyle:** The Lifestyle family shows a **fairly balanced distribution** between the **postnatal** (53.4%) and **prenatal** (46.6%) periods. This indicates that lifestyle factors influence the phenotype both during and after pregnancy, with a slight emphasis on the postnatal period.

- **Natural Spaces:** Interestingly, the Natural Spaces family is exclusively influenced postnatally, with a 100% contribution. This highlights the importance of environmental factors such as green spaces on the phenotype after birth.

4 Discussion

To check the coherence of our results we did some research on the existing literature. We found sources supporting the following arguments :

- A study by David C. Bellinger (2013) [1] demonstrated that high levels of heavy metals such as lead and manganese, in prenatal or early childhood blood, were linked with cognitive deficits and behavioral issues.
- A study by Petrican, Fornito and Boyland (2024) [4] on adolescents showed that a healthy diet has positive psychological outcomes (i.e., superior attentional skills and interpersonal outcomes). Given that diet is a key determinant of gut microbiota that are strongly implicated in the regulation of brain activity and cognitive function.
- A study by Dockx et al. (2022) [2] found that early exposure to green spaces is associated with reduced hyperactive behavior, better attention-related outcomes and better visual memory-related outcomes.

In conclusion, to mitigate the negative effects of harmful exposures and promote healthier neurobehavioral development in children, there are a few things we should be careful about from the prenatal to the postnatal period. The main points of focus, that we saw through our statistical analysis should be : prenatal and postnatal exposure to metals, prenatal and postnatal lifestyle (more specifically the diet), and exposure to natural spaces.

Avoiding metals may involve being in contact with as few as possible products or environments with high levels of lead or other heavy metals (old paints, plumbing), ensure safe drinking water by maybe considering testing it for contaminants.

Including probiotics and prebiotics in our daily diet promotes healthy gut microbiota, which is known to positively influence brain function.

Encouraging children to spend more time in parks, gardens, and other natural settings will make them benefit from the cognitive and behavioral advantages associated with green spaces.

References

- [1] David C. Bellinger. Prenatal exposures to environmental chemicals and children’s neurodevelopment: An update. *Safety and Health at Work*, 4(1):1–11, 2013.
- [2] Yinthe Dockx, Esmée M. Bijmens, Leen Luyten, Martien Peusens, Eline Provost, Leen Rasking, Hanne Sleurs, Janneke Hogervorst, Michelle Plusquin, Lidia Casas, and Tim S. Nawrot. Early life exposure to residential green space impacts cognitive functioning in children aged 4 to 6 years. *Environment International*, 161:107094, 2022.
- [3] Léa Maitre, Jean-Baptiste Guimbaud, Charline Warembourg, Nuria Güil-Oumrait, Paula Marcela Petrone, Marc Chadeau-Hyam, Martine Vrijheid, Xavier Basagaña, and Juan R. Gonzalez. State-of-the-art methods for exposure-health studies: Results from the exposome data challenge event. *Environment International*, 168:107422, 2022.
- [4] Raluca Petrican, Alex Fornito, and Emma Boyland. Lifestyle factors counteract the neurodevelopmental impact of genetic risk for accelerated brain aging in adolescence. *Biological Psychiatry*, 95(5):453–464, 2024. BPS 95/5Mood Disorder Biomarkers.